

Tianchu Zeng

PhD Candidate, Electrical & Computer Engineering · National University of Singapore

+65 8790 5814 · tianchuzeng@gmail.com

ORCID: [0009-0009-0773-4502](https://orcid.org/0009-0009-0773-4502) · [Google Scholar](#) · [GitHub](#) · [LinkedIn](#) · [Website](#)

PROFESSIONAL SUMMARY

Machine-learning researcher and NUS PhD candidate developing efficient scientific models and reliable evaluation methods. Co-first author of a study accepted in principle at Nature Methods, accelerating biophysical brain-model fitting by 50-100× across a study of 12,005 participants; developer of SHARP for algorithm comparison with dependent cross-validation estimates.

RESEARCH INTERESTS

Computational neuroscience · AI for science · statistical methods for machine-learning evaluation · biophysical brain modeling · population-scale neuroimaging · scientific and brain foundation models

EDUCATION

National University of Singapore — Singapore

Aug 2022 - Present (expected 2027)

Ph.D. in Electrical & Computer Engineering · Advisors: BT Thomas Yeo and Thomas E. Nichols · GPA: 4.88/5.0

Tsinghua University — Beijing, China

Aug 2018 - Jun 2022

B.Eng. in Department of Automation · GPA: 3.74 / 4.00

RESEARCH PORTFOLIO

DELSSOME — Deep-learning surrogates for biophysical brain modeling

Aug 2022 - Present

- Designed and trained deep-learning surrogates and integrated them into simulation-validated optimization, accelerating biophysical brain-model fitting by 50-100×.
- Coordinated individual-level evaluation across 12,005 participants from 14 datasets, yielding model-derived normative lifespan trajectories of cortical excitation-inhibition balance.

SHARP — Valid inference for repeated cross-validation

2024 - Present

- Developed SHARP, a split-half repeated cross-validation method for comparing learning algorithms while accounting for dependence; evaluated false-positive control, power, and confidence-interval calibration across 420 scenarios.
- Led a PRISMA-guided review of 210 biomedical machine-learning studies; 97% of assessed studies ignored fold dependence in comparisons of predictive performance.

Brain Foundation Models — Rigorous benchmarking for phenotypic prediction

2025 - Present

- Fine-tunes brain foundation models and designs evaluations comparing their phenotypic-prediction performance with classical methods.

ADDITIONAL RESEARCH

Brain Parcellation with Resting-State fMRI — National University of Singapore

Aug 2022 - Jun 2023

- Developed group- and individual-level parcellation methods using functional-connectivity analysis and cortical-surface visualization.

Multi-Agent Trajectory Prediction — Tsinghua University

Jan 2020 - Jun 2022

- Built RNN-, GNN-, CVAE-, and Transformer-based models; the work produced two co-first-authored peer-reviewed conference papers and one granted Chinese patent.

SELECTED PUBLICATIONS

* denotes equal contribution and co-first authorship; + denotes co-corresponding authorship; author name in bold.

1. **Zeng, T.***, Tian, F.* , Zhang, S., Li, X., Tan, A. P., Larsen, B., ... & Yeo, B. T. T. (2026). Optimizing biophysical large-scale brain circuit models with deep neural networks. *Nature Methods*. Accepted in principle. [Preprint](#).
2. **Zeng, T.***, Li, H.* , Zhang, S.* , Tan, Y. Q., Tian, F., Orban, C., ..., Nichols, T.+ , & Yeo, B. T.+ (2026). Widespread use of invalid statistical tests in biomedical machine learning. *bioRxiv* preprint. [DOI](#).
3. Qu, Y. L., Chen, J., Tam, A., Ooi, L. Q. R., Dhamala, E., Cocuzza, C. V., Zhang, S., **Zeng, T.**, ... & Holmes, A. J. (2025). Distinct brain network features predict internalizing and externalizing traits in children, adolescents and adults. *Nature Mental Health*, 3(3), 306-317.
4. Zhang, S.* , Larsen, B.* , Sydnor, V. J.* , **Zeng, T.**, An, L., Yan, X., ... & Yeo, B. T. (2024). In vivo whole-cortex marker of excitation-inhibition ratio indexes cortical maturation and cognitive ability in youth. *Proceedings of the National Academy of Sciences*, 121(23), e231864121.
5. **Zeng, T.**, & Yeo, B. T. T. (2026). Deep learning charts the excitation-inhibition balance of the brain throughout the lifespan. *Nature Methods*. Invited Research Briefing accompanying the DELSSOME study. Accepted.

SCIENTIFIC SOFTWARE AND OPEN SOURCE

[DELSSOME](#) — Lead Developer

- Leads the official implementation for the Nature Methods study, covering group- and individual-level pipelines, pretrained models, reproducible examples, documentation, and regression tests.
- Built an end-user workflow that estimates cortical excitation-inhibition ratios from functional-connectivity and FCD inputs while keeping validation tied to numerical simulation.

[ThomasYeoLab CBIG](#) — Research Software Contributor

- Authored and maintains the Zeng2026_DELSSOME stable project; reviewed the pull request for Zhang2024_pFIC.

[admissions-ops](#) — Creator and Maintainer

- Built a local-first AI-agent workspace that generates sourced evaluations and validates required fields, provenance, file structure, and human-review checkpoints.

[career-ops](#) — Contributor

- Contributes to the career-ops open-source project.

INDUSTRY EXPERIENCE

R&D Engineer Intern — Beijing Zhangshangxianji Network Technology Co., Ltd.

Jun 2021 - Sep 2021

- Implemented Spring Boot transaction services and unified Single Sign-On; built SMS authentication, OCR-resistant verification, complex database access, and department-wide messaging and automated-call alerts.

SELECTED HONORS AND ACTIVITIES

- Best Poster Award (Basic Science), NUHS Scientific and Innovation Summit, 2026.
- NUS Research Scholarship, 2022-2026.
- Invited talk at the Asia-Pacific Computational and Cognitive Neuroscience Conference, RIKEN, 2025
- Posters at OHBM 2026, NUHS 2026, and AI4X 2025.
- Teaching Assistant for EE2026 Digital Design and EE3731C Signal Processing at NUS.

TECHNICAL SKILLS

- **Machine learning and scientific computing:** Python, MATLAB, PyTorch, JAX, scikit-learn, SciPy; deep learning, surrogate modeling, numerical optimization, and simulation
- **Statistics:** cross-validation inference, hypothesis testing, simulation-based validation, false-positive-rate, power, and confidence-interval calibration
- **Neuroimaging:** resting-state fMRI, functional connectivity, brain parcellation, cortical-surface visualization, FreeSurfer/Freeview
- **Research engineering:** Git, GitHub Actions, Linux/bash, HPC/PBS, conda, Docker; reproducible workflows, provenance, and automated validation